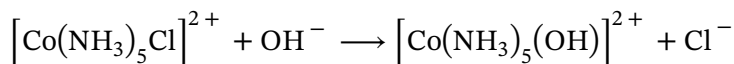


The cobalt(III) pentammine complex, $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$, consists of a central cobalt cation with coordinate covalent bonds to five neutral ammonia molecules and a chloride anion, which act as ligands. In an aqueous solution with a pH > 10, the complex readily undergoes a ligand exchange via base hydrolysis as shown in Reaction 1.



Reaction 1

Researchers have proposed that Reaction 1 might proceed by either a bimolecular nucleophilic substitution (an $\text{S}_{\text{N}}2$ mechanism) as shown in Figure 1, or by a unimolecular nucleophilic substitution of the complex's conjugate base (an $\text{S}_{\text{N}}1\text{CB}$ mechanism) as shown in Figure 2.

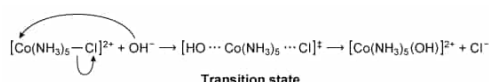


Figure 1 $\text{S}_{\text{N}}2$ mechanism proposed for Reaction 1

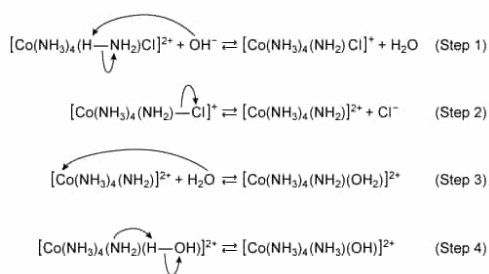


Figure 2 $\text{S}_{\text{N}}1\text{CB}$ mechanism proposed for Reaction 1

In the proposed $\text{S}_{\text{N}}2$ mechanism, the OH^- ion functions as a nucleophile that attacks the central Co atom in the complex. Accordingly, the nucleophilic attack results in a transition state in which the existing Co–Cl bond breaks and a new Co–OH bond forms in a single step.

In contrast, the $\text{S}_{\text{N}}1\text{CB}$ mechanism proposes that the reaction occurs in multiple steps in which OH^- initially acts as a base to deprotonate one of the coordinately bonded ammine ligands to form a hexa-coordinated intermediate with an amido (NH_2^-) ligand (Step 1). The Cl^- ligand in the amido intermediate then dissociates to form a pentacoordinated intermediate (Step 2). A water molecule from the solution then coordinates with the Co atom of the pentacoordinated intermediate to form an aquo complex (Step 3). A solvent-mediated proton transfer from the aquo to the amido ligand yields the final base-hydrolysis product (Step 4).

In the $\text{S}_{\text{N}}1\text{CB}$ mechanism, what is the most likely molecular geometry around the central Co atom in the

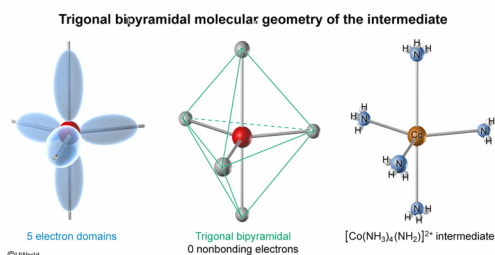
$[\text{Co}(\text{NH}_3)_4(\text{NH}_2)]^{2+}$ intermediate?

- ☐ A. Tetrahedral [9%]
- ☐ B. Octahedral [22%]
- ☒ C. Trigonal bipyramidal [51%]
- ☐ D. Square pyramidal [16%]

Correct

50% answered correctly

Explanation:



According to valence-shell electron-pair repulsion (VSEPR) theory, the electron pairs around a central atom in a molecule will adopt an **electron geometry** (ie, a spatial arrangement) that maximizes the separation between the electron domains and minimizes the repulsion between the electrons. Using VSEPR theory, the **molecular geometry** (ie, the shape) of a molecule can be predicted from its Lewis structure by evaluating the orientations that the *bonded atoms* will adopt due to the number of bonding and nonbonding electron domains present around the central atom.

The $[\text{Co}(\text{NH}_3)_4(\text{NH}_2)]^{2+}$ intermediate formed in Step 2 of the $\text{S}_{\text{N}}1\text{CB}$ mechanism (Figure 2) consists of a central cobalt cation surrounded by five electron domains (ie, four coordinate covalent bonds to NH_3 molecules, one coordinate covalent bond to an NH_2^- anion, and no nonbonding electron pairs). As such, VSEPR theory predicts a **trigonal bipyramidal** electron geometry, and because all the electron domains around the Co atom are bonding electrons (ie, no nonbonding electron pairs), the molecular geometry is also trigonal bipyramidal.

(Choice A) A **tetrahedral** molecular geometry would result if there were only four coordinate bonds and no nonbonding electron pairs around the central Co atom.

(Choice B) An **octahedral** molecular geometry would result if there were six electron domains around the Co atom (eg, six ligand bonding domains and no nonbonding electron pairs).

(Choice D) A **square pyramidal** molecular geometry would result if the Co atom had five coordinate bonds and one nonbonding electron pair (ie, six electron domains), but the $[\text{Co}(\text{NH}_3)_4(\text{NH}_2)]^{2+}$ intermediate has only five ligand-bonding domains.

Educational objective:

Valence-shell electron-pair repulsion (VSEPR) theory holds that the electron domains around a central atom will adopt an electron geometry that maximizes the separation and minimizes the repulsion between the electrons. The molecular geometry (shape) of a molecule depends on the electron geometry and number of nonbonding electron pairs of the central atom.

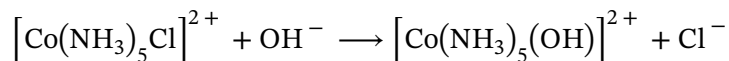
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Subject:
General Chemistry

QID: 404077

System:
Interactions of Chemical Substances

The cobalt(III) pentammine complex, $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$, consists of a central cobalt cation with coordinate covalent bonds to five neutral ammonia molecules and a chloride anion, which act as ligands. In an aqueous solution with a pH > 10, the complex readily undergoes a ligand exchange via base hydrolysis as shown in Reaction 1.



Reaction 1

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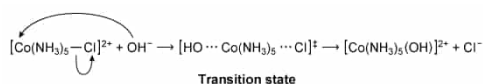


Figure 1 $\text{S}_{\text{N}}2$ mechanism proposed for Reaction 1

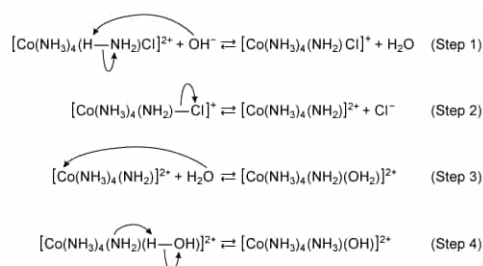


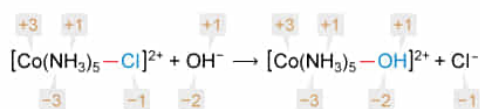
Figure 2 $\text{S}_{\text{N}}1\text{CB}$ mechanism proposed for Reaction 1

In the proposed $\text{S}_{\text{N}}2$ mechanism, the OH^- ion functions as a nucleophile that attacks the central Co atom in the complex. Accordingly, the nucleophilic attack results in a transition state in which the existing Co–Cl bond breaks and a new Co–OH bond forms in a single step.

In contrast, the $\text{S}_{\text{N}}1\text{CB}$ mechanism proposes that the reaction occurs in multiple steps in which OH^- initially acts as a base to deprotonate one of the coordinately bonded ammine ligands to form a hexacoordinated intermediate with an amido (NH_2^-) ligand (Step 1). The Cl^- ligand in the amido intermediate then dissociates to form a pentacoordinated intermediate (Step 2). A water molecule from the solution then coordinates with the Co atom of the pentacoordinated intermediate to form an aquo complex (Step 3). A solvent-mediated proton transfer from the aquo to the amido ligand yields the final base-hydrolysis product (Step 4).

Is the ligand exchange in Reaction 1 an oxidation–reduction reaction?

- ☐ A. Yes, because the Co atom gains two electrons in the reaction. [6%]
- ☐ B. Yes, because the Co atom loses two electrons in the reaction. [7%]
- ☐ C. Yes, because the net charge of the complex decreases from +2 to +1 in the first step of the mechanism. [10%]
- ☒ D. No, because the oxidation numbers of all atoms remain unchanged during the reaction. [75%]

Explanation:**Ligand exchange is not a redox reaction**

Ligands exchange via a coordinate bond,
but the oxidation numbers are unchanged.

In an **oxidation–reduction (redox) reaction**, an atom is **reduced** by gaining one or more electrons from another atom that loses the electrons (ie, gets **oxidized**). To track the gain or loss of electrons by atoms in reactions, **oxidation numbers** can be assessed **using oxidation state rules** or can be calculated **using Lewis structures**. **Comparing the oxidation number** of an element in the reactants with the oxidation number of that same element in the products indicates whether the element has gained or lost electrons.

The breaking or forming of **coordinate bonds** is not a redox reaction. Coordinate bonds form by a **Lewis acid–base** interaction in which a **lone pair of electrons** occupying an orbital of an **electron-rich atom** in the ligand (ie, the Lewis base) "plugs into" a vacant orbital of an **electron-deficient metal** cation (ie, the Lewis acid) and the electrons are shared. As such, ligands retain possession of the shared electrons and can "unplug" from a complex and be exchanged for other ligands.

In Reaction 1, a coordinate bond to a Cl^- ligand is exchanged for a coordinate bond to a OH^- ligand, and the Cl and O atoms are both more electronegative than Co. As a result, the Co atom has no net gain or loss of electrons and no change in its oxidation state of +3. The oxidation numbers of all other atoms also remain unchanged. Therefore, Reaction 1 is not an oxidation–reduction reaction.

(Choices A and B) The Co atom has no net gain or loss of electrons in Reaction 1.

(Choice C) The net formal charge of the complex does change in the first step of the mechanism, but this is due to the transfer of a proton (ie, an H^+ ion), not due to the gain or loss of electrons.

Educational objective:

Oxidation numbers track the transfer of electrons between atoms during a redox reaction. An increased oxidation number indicates a loss of electrons (oxidation) whereas a decreased oxidation number results from gaining electrons (reduction). A change of one unit occurs for each electron transferred.

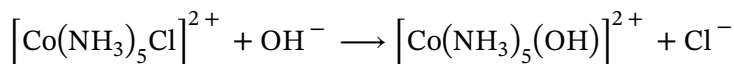
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Subject:

General Chemistry

QID: 404078
System:

Interactions of Chemical Substances

The cobalt(III) pentammine complex, $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$, consists of a central cobalt cation with coordinate covalent bonds to five neutral ammonia molecules and a chloride anion, which act as ligands. In an aqueous solution with a pH > 10, the complex readily undergoes a ligand exchange via base hydrolysis as shown in Reaction 1.



Reaction 1

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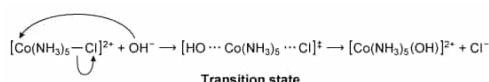


Figure 1 $\text{S}_{\text{N}}2$ mechanism proposed for Reaction 1

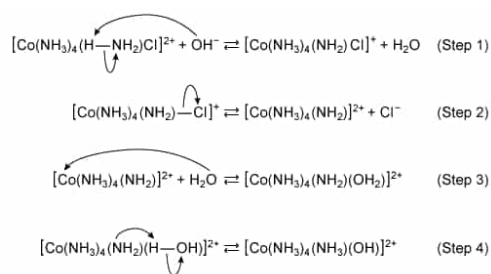


Figure 2 $\text{S}_{\text{N}}1\text{CB}$ mechanism proposed for Reaction 1

In the proposed $\text{S}_{\text{N}}2$ mechanism, the OH^- ion functions as a nucleophile that attacks the central Co atom in the complex. Accordingly, the nucleophilic attack results in a transition state in which the existing Co–Cl bond breaks and a new Co–OH bond forms in a single step.

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What effect will the ligand exchange (Reaction 1) have on the pH of the solution? The pH will:

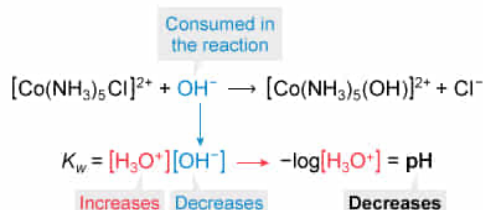
- ✓ ☒ A. decrease slightly. [65%]
☐ B. not change. [12%]
☐ C. increase slightly. [19%]
☐ D. increase by 1 unit. [1%]

Correct

65% answered correctly

Explanation:

Change in pH as Reaction 1 proceeds



The pH of a solution relates directly to the hydrogen ion concentration $[\text{H}^+]$ (which is sometimes expressed in the form of the **hydronium ion** concentration, $[\text{H}_3\text{O}^+]$). As such, the **pH** of a solution is given by the relationship:

$$\text{pH} = -\log[\text{H}_3\text{O}^+] = -\log[\text{H}^+]$$

If the hydroxide concentration $[\text{OH}^-]$ is known instead, it is related to $[\text{H}_3\text{O}^+]$ by the self-ionization constant of water K_w , which is constant at a given temperature. At 25 °C:

$$K_w = [\text{H}_3\text{O}^+][\text{OH}^-] = 1.0 \times 10^{-14}$$

In Reaction 1, the ligand exchange involves OH^- as a reactant (ie, OH^- is bound to the complex in place of Cl^-). As such, the $[\text{OH}^-]$ decreases as the reaction proceeds. The K_w expression shows that as **$[\text{OH}^-]$ decreases, $[\text{H}_3\text{O}^+]$ increases** (and vice versa):

$$K_w = [\text{increasing } \text{H}_3\text{O}^+][\text{decreasing } \text{OH}^-] = 1.0 \times 10^{-14} \text{ (constant at } 25^\circ\text{C)}$$

Accordingly, the resulting increase in $[\text{H}_3\text{O}^+]$ corresponds to a decrease in pH because the **pH** is the negative logarithm of concentration (ie, a **higher $[\text{H}_3\text{O}^+]$ yields a lower pH**):

$$\text{decreasing pH} = -\log[\text{increasing } \text{H}_3\text{O}^+]$$

Although the extent of the change in Reaction 1 depends on how much of the $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ complex reacts, the reaction will cause the pH to decrease slightly.

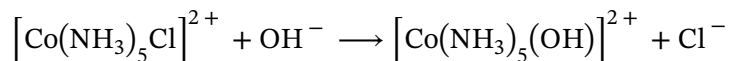
(Choice B) The pH must *decrease* proportionally to the amount of complex that reacts because OH^- is a reactant and $[\text{OH}^-]$ decreases as the reaction proceeds.

(Choices C and D) As $[\text{OH}^-]$ decreases due to the reaction, $[\text{H}_3\text{O}^+]$ will increase due to the definition of K_w . The pH will *decrease* as $[\text{H}_3\text{O}^+]$ increases.

Educational objective:

The $[\text{H}_3\text{O}^+]$ and $[\text{OH}^-]$ concentrations are related by the self-ionization (autoionization) constant of water (K_w) according to the expression $K_w = [\text{H}_3\text{O}^+][\text{OH}^-]$, which is constant at a given temperature (equals 1.0×10^{-14} at 25 °C). As $[\text{OH}^-]$ decreases, $[\text{H}_3\text{O}^+]$ increases (and vice versa). The pH is related to $[\text{H}_3\text{O}^+]$ by $\text{pH} = -\log[\text{H}_3\text{O}^+]$.

The cobalt(III) pentammine complex, $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$, consists of a central cobalt cation with coordinate covalent bonds to five neutral ammonia molecules and a chloride anion, which act as ligands. In an aqueous solution with a pH > 10, the complex readily undergoes a ligand exchange via base hydrolysis as shown in Reaction 1.



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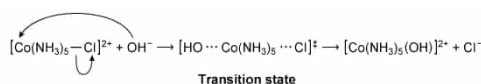


Figure 1 $\text{S}_{\text{N}}2$ mechanism proposed for Reaction 1

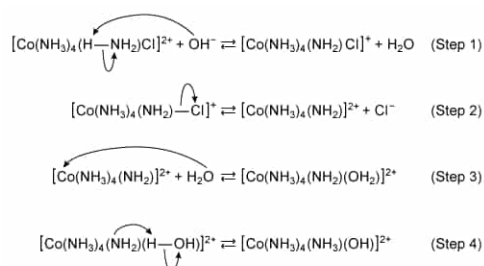


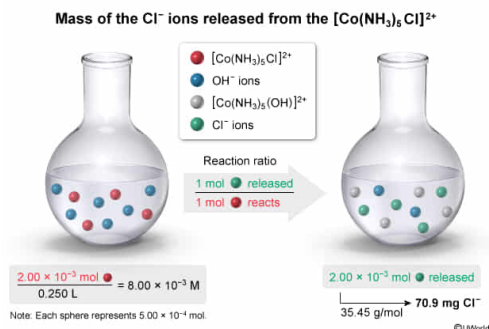
Figure 2 $\text{S}_{\text{N}}1\text{CB}$ mechanism proposed for Reaction 1

In the proposed $\text{S}_{\text{N}}2$ mechanism, the OH^- ion functions as a nucleophile that attacks the central Co atom in the complex. Accordingly, the nucleophilic attack results in a transition state in which the existing Co–Cl bond breaks and a new Co–OH bond forms in a single step.

In contrast, the $\text{S}_{\text{N}}1\text{CB}$ mechanism proposes that the reaction occurs in multiple steps in which OH^- initially acts as a base to deprotonate one of the coordinately bonded ammine ligands to form a hexa-coordinated intermediate with an amido (NH_2^-) ligand (Step 1). The Cl^- ligand in the amido intermediate then dissociates to form a pentacoordinated intermediate (Step 2). A water molecule from the solution then coordinates with the Co atom of the pentacoordinated intermediate to form an aquo complex (Step 3). A solvent-mediated proton transfer from the aquo to the amido ligand yields the final base-hydrolysis product (Step 4).

If 250 mL of an $8.00 \times 10^{-3} \text{ M}$ $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}(\text{aq})$ solution undergoes Reaction 1, what is the mass of the Cl^- ions released from the complex? (Note: Assume that OH^- is in excess and that the reaction goes to completion.)

- ☐ A. 0.28 mg [6%]
- ☐ B. 7.10 mg [21%]
- ☒ C. 71.0 mg [68%]
- ☐ D. 142 mg [3%]

Explanation:

The **molar concentration (molarity)** of a chemical species is defined as the **number of moles** of the species **per liter of solution (mol/L)**. As such, if a solution has a concentration of $8.00 \times 10^{-3} \text{ M}$ $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$, the number of moles of $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ present in 250 mL (0.250 L) is:

$$0.250 \text{ L solution} \times \frac{8.00 \times 10^{-3} \text{ mol} [\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}}{1 \text{ L solution}} = 2.00 \times 10^{-3} \text{ mol} [\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$$

The question specifies that OH^- is in excess; therefore, $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ is the **limiting reactant** (ie, the reactant that gets used up first) in the reaction mixture and determines the amount of product that can be formed. The **mole ratio** (stoichiometry) of $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ to Cl^- in Reaction 1 indicates that 1 mole of Cl^- is released for every 1 mole of $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ that reacts. Accordingly, if the reaction goes to completion, the number of moles of Cl^- ions released from the complex in the reaction is:

$$2.00 \times 10^{-3} \text{ mol} [\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+} \times \frac{1 \text{ mol Cl}^- \text{ released}}{1 \text{ mol} [\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}} = 2.00 \times 10^{-3} \text{ mol Cl}^- \text{ released}$$

Multiplying the moles of Cl^- by its **molar mass** from the periodic table yields:

$$2.00 \times 10^{-3} \text{ mol Cl}^- \times \frac{35.5 \text{ g Cl}^-}{1 \text{ mol Cl}^-} = 71.0 \times 10^{-3} \text{ g Cl}^- \Rightarrow 71.0 \text{ mg Cl}^-$$

(Choice A) A value of 0.28 (with incorrect units) is obtained if the molarity of the $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ is taken as the number of moles (ie, the molarity of the solution is not multiplied by its volume in liters).

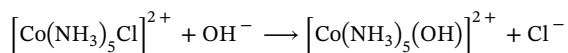
(Choice B) A value of 7.10 (with incorrect units) is obtained if a math error is made when converting the solution volume from milliliters to liters in the first step of the calculation.

(Choice D) A result of 142 mg is found if the molar mass of Cl_2 is used instead of the mass of Cl^- in the last step of the calculation.

Educational objective:

The number of moles of a species in a solution can be calculated by multiplying the molarity of the solution by its volume. Mole ratios from a balanced reaction relate the moles of one species to the moles of another species, and the mass of a given number of moles of a species can be found using its molar mass.

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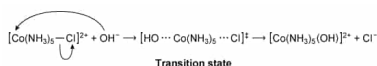


Figure 1 $\text{S}_{\text{N}}2$ mechanism proposed for Reaction 1

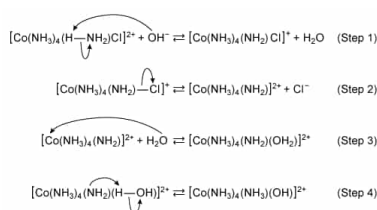
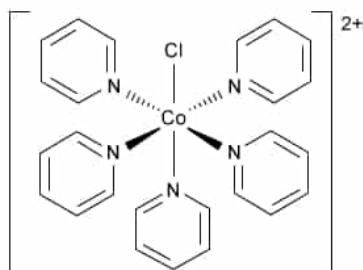


Figure 2 $\text{S}_{\text{N}}1\text{CB}$ mechanism proposed for Reaction 1

In the proposed $\text{S}_{\text{N}}2$ mechanism, the OH^- ion functions as a nucleophile that attacks the central Co atom in the complex. Accordingly, the nucleophilic attack results in a transition state in which the existing Co–Cl bond breaks and a new Co–OH bond forms in a single step.

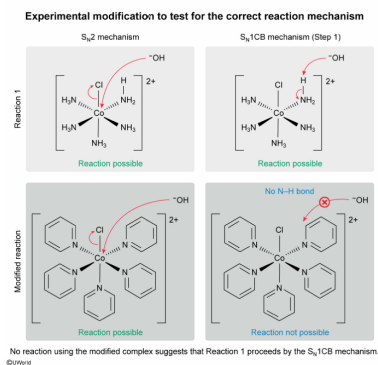
In contrast, the $\text{S}_{\text{N}}1\text{CB}$ mechanism proposes that the reaction occurs in multiple steps in which OH^- initially acts as a base to deprotonate one of the coordinately bonded ammine ligands to form a hexa-coordinated intermediate with an amido (NH_2^-) ligand (Step 1). The Cl^- ligand in the amido intermediate then dissociates to form a pentacoordinated intermediate (Step 2). A water molecule from the solution then coordinates with the Co atom of the pentacoordinated intermediate to form an aquo complex (Step 3). A solvent-mediated proton transfer from the aquo to the amido ligand yields the final base-hydrolysis product (Step 4).



To evaluate the proposed mechanisms, researchers attempted to modify Reaction 1 by replacing the $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ complex with the complex shown above, which contains $\text{C}_5\text{H}_5\text{N}$ (pyridine) ligands instead of NH_3 ligands, but no reaction occurred. Assuming that steric hindrance is not a factor in the reaction, this result provides evidence in support of the:

- ☐ A. $\text{S}_{\text{N}}2$ mechanism, because the experimental change resulted in no reaction. [19%]
- ☐ B. $\text{S}_{\text{N}}2$ mechanism, because the pyridine ligands have a negative formal charge that repels the OH^- ion. [9%]
- ☐ C. $\text{S}_{\text{N}}1\text{CB}$ mechanism, because the leaving group is the same in both the original and the modified reaction. [15%]
- ☒ D. $\text{S}_{\text{N}}1\text{CB}$ mechanism, because an N–H bond is required to form the proposed intermediate. [56%]

Explanation:



A **reaction mechanism** is a sequence of **elementary reaction steps** that yield an **overall (net) reaction**. Mechanisms include the reactants, intermediates, products, and catalysts involved in each step. A proposed mechanism must be experimentally tested to determine whether it is correct.

Experimental methods to test proposed mechanisms may include:

- **Verifying the reaction's rate law:** If a reaction follows a proposed mechanism, the rate of the reaction will change according to the mechanism's **rate law** as the initial concentrations of the reactants are changed.
- **Detecting an intermediate:** If a reaction follows a mechanism that forms an intermediate, adding a substance that reacts only with the intermediate would form a side product that is detectable by instrumental methods (eg, **spectroscopy**). If no side product is detected, the intermediate probably does not form, and the proposed mechanism is likely incorrect.
- **Interfering with a mechanistic pathway:** If there is a unique feature in one mechanism that is not present in another mechanism, an experiment can be modified in a way that would interfere with one mechanism but would not impact the other mechanism.

Of the two mechanisms described in the passage, only the S_N1CB mechanism forms an intermediate by the **deprotonation** of an N–H bond. Therefore, replacing the $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ complex in Reaction 1 with a complex that has $\text{C}_5\text{H}_5\text{N}$ (pyridine) ligands without N–H bonds should interfere with the S_N1CB mechanism and alter the outcome if Reaction 1 occurs by that pathway.

(Choice A) Although there was no reaction, the experimental change responsible for this result was not relevant to the pathway of Mechanism 1 and consequently does not provide any evidence to support it.

(Choice B) Both the NH_3 ligands and the $\text{C}_5\text{H}_5\text{N}$ ligands are neutral molecules without a net formal charge.

(Choice C) Having the same leaving group does not provide a distinguishing difference to discriminate between the two mechanisms.

Educational objective:

A reaction mechanism is a sequence of elementary reaction steps that yield an overall reaction. Experimental methods to test a proposed mechanism may include verifying that the reaction's rate law is consistent with a proposed mechanism, detecting proposed intermediates, and performing experiments that could interfere with distinguishing steps in a possible mechanistic pathway.

Time spent: 0 sec

Subject:
General Chemistry

QID: 404081

System:
Interactions of Chemical Substances